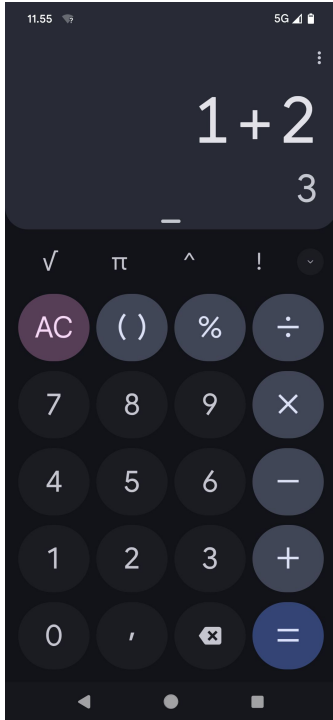


Kodning

C#

Nyt projekt: Lommeregner

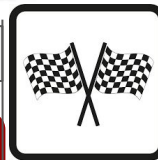


Plan

Til og med 7. Marts



Uge 3	Uge 4	Uge 5	Uge 6	Uge 7	Uge 8	Uge 9	Uge 10
Feature 1: Lav UI							
		Feature 2: Få knapper til at printe tal					
				Feature 3: Lav metoder til + og -			
						Feature 4: Freestyle	

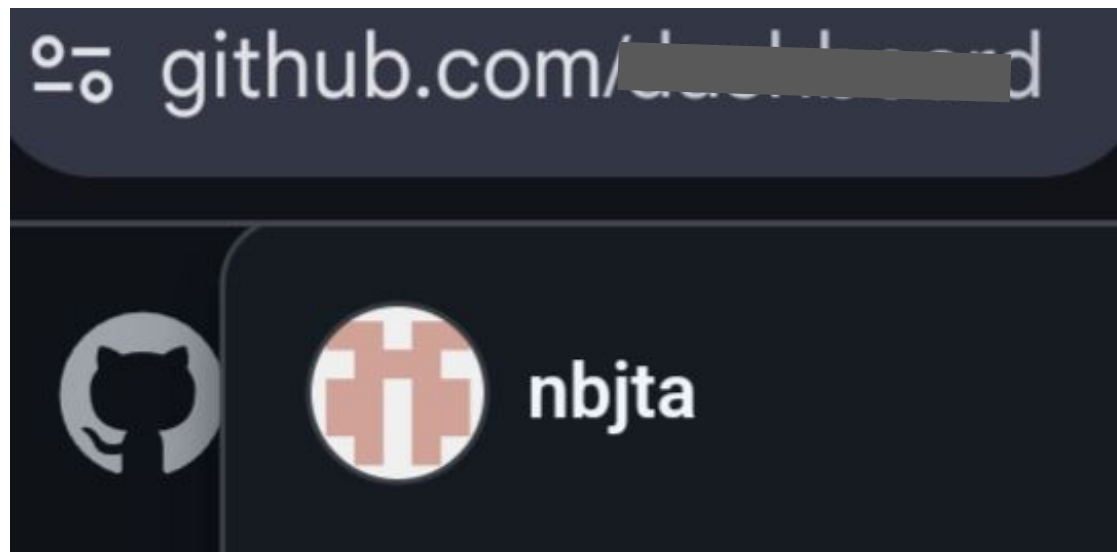




Feature 1: Lav UI

Find mig på github.com

- Gå til github.com
- Søg på nbjta
- Download Feature1.pdf



Calculator

The screenshot shows the Visual Studio IDE with a project named **WinFormsApp1**. The code editor displays the following C# code in **Form1.cs**:

```
1 namespace WinFormsApp1
2 {
3     public partial class Form1 : Form
4     {
5         public Form1()
6         {
7             InitializeComponent();
8         }
9     }
10 }
```

The Solution Explorer on the right shows the project structure:

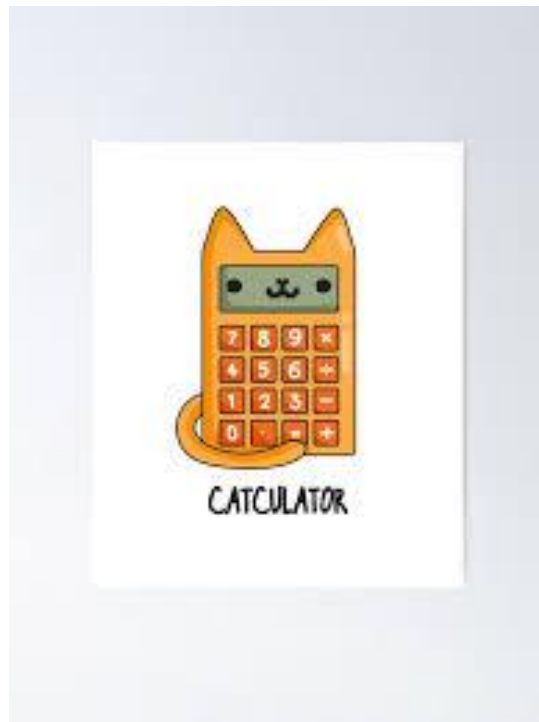
- Solution: WinFormsApp1 (1 of 1 project)
- WinFormsApp1
- Dependencies
- Form1.cs
- C# Program.cs

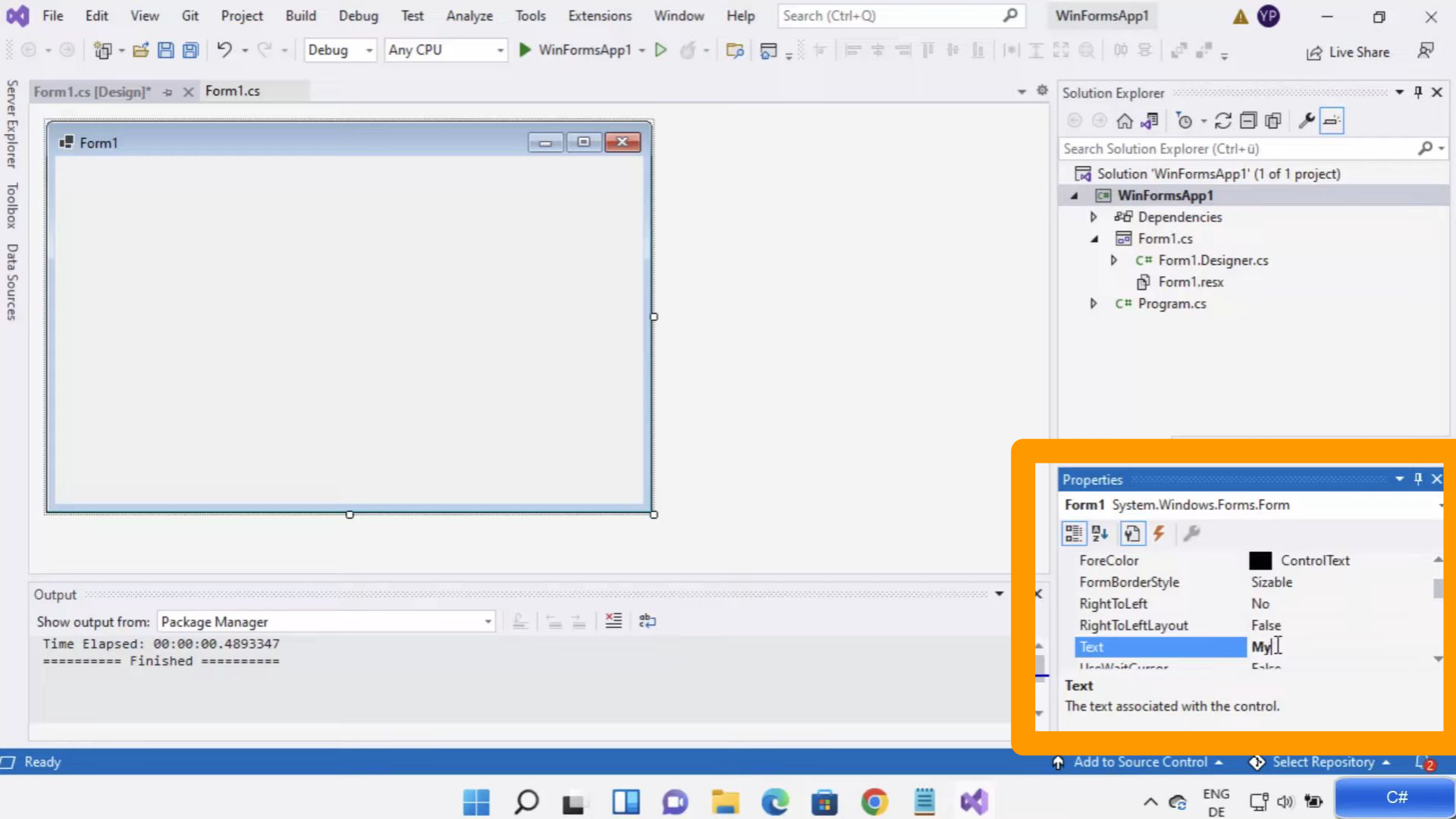
The Output window at the bottom shows the package manager restoring the project:

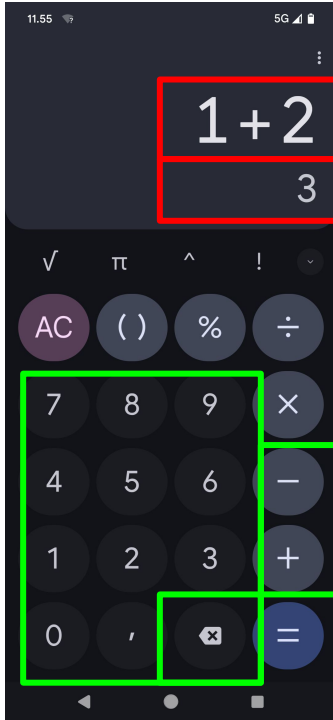
```
Show output from: Package Manager
Restored C:\Users\codebind\source\repos\WinFormsApp1\WinFormsApp1\WinFormsApp1.csproj (in 2.62 sec).
Time Elapsed: 00:00:09.1100467
===== Finished =====
```

Red boxes highlight the project name in the Solution Explorer, the namespace in the code, and the class name in the code. Red lines connect the 'Calculator' title to these elements.

Styling







To do:

Tekstfelt til regnestykke

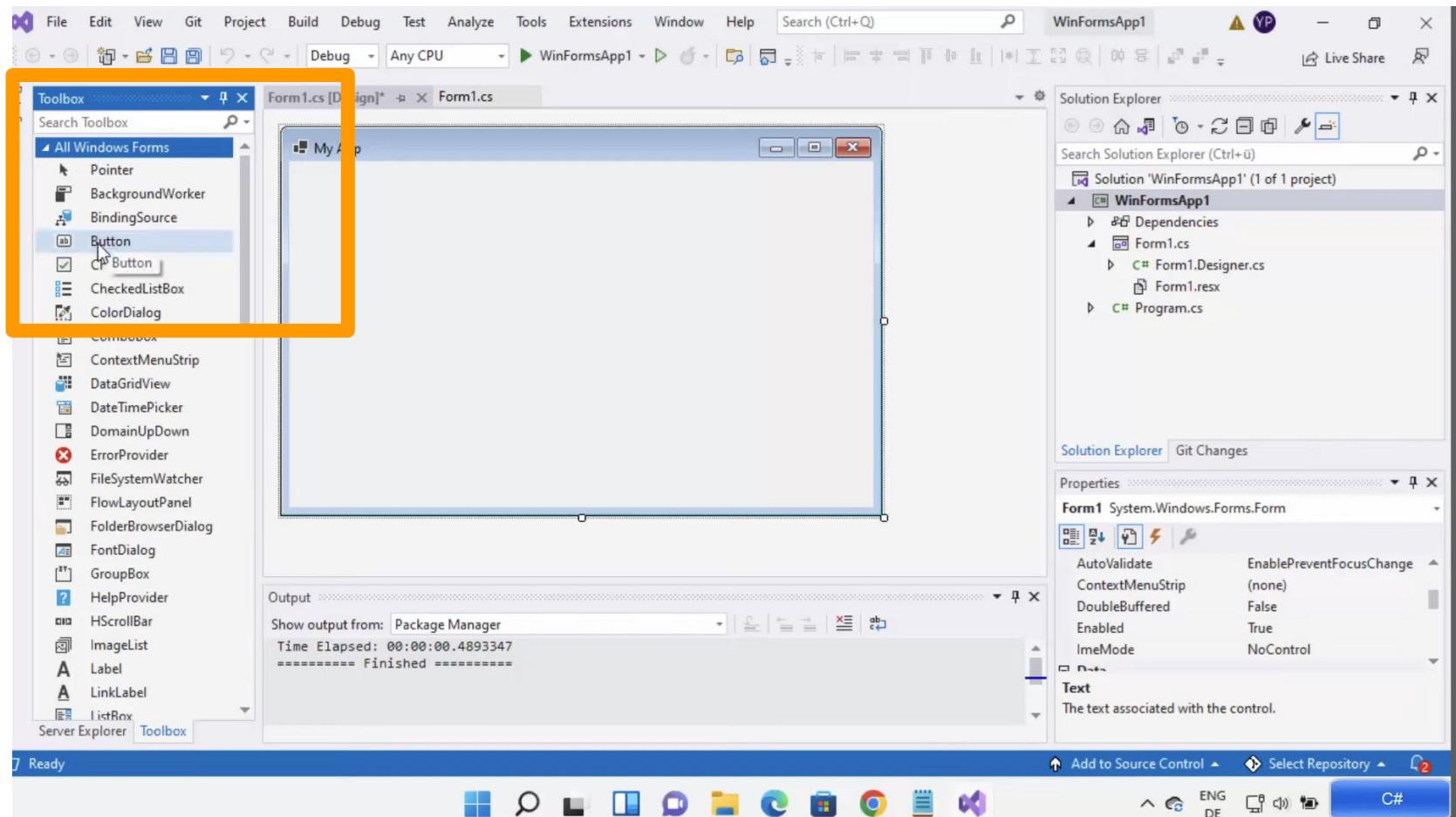
Tekstfelt til resultat

Knapper med tal

Knap til at slette sidste tal

Knapper til + og -

Klik på View og vælg Toolbox i top menu'en.



God kode er
let at forstå for andre

- A) “No magic numbers”
- B) God navngivning
- C) Minimum gentagelser

Derfor:

- Knapper og tekstfelter skal have gode navne

ButtonAddition

ButtonOne

ButtonSubtract

ButtonTwo

ButtonThree

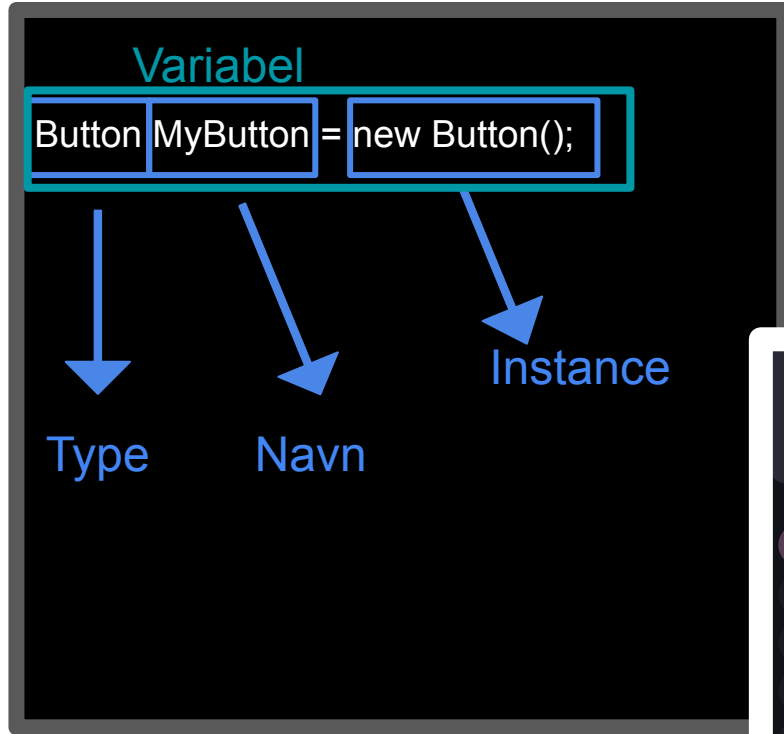
TexviewFormula

TextviewResult

Pro tip:
Et godt navn
fortæller hvad
knappen bliver
brugt til.



Dårlig navngivning

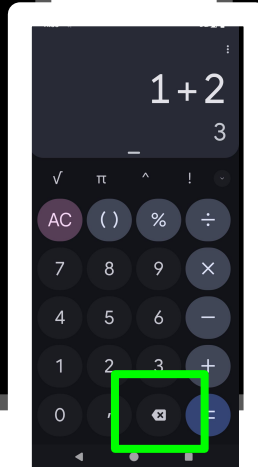


VS

God navngivning

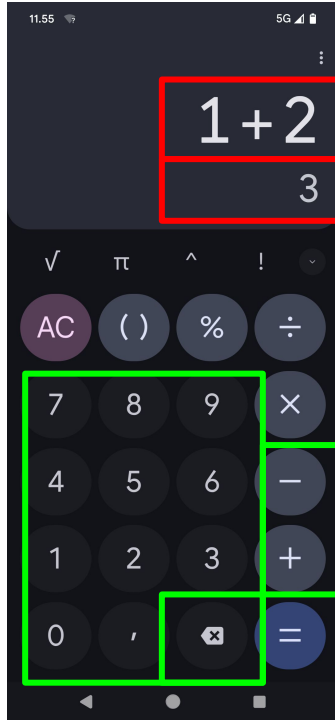


```
Button DeleteButton = new Button();
```

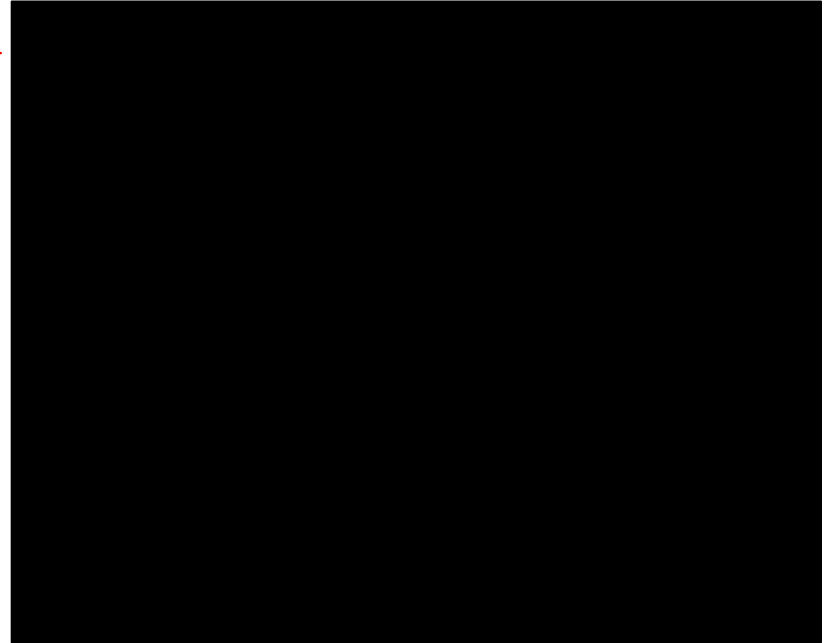


Find 5 fejl

Feature 2: Få knapper til at printe tal



To do:



Data Types

bool	Boolean value
byte	8-bit unsigned integer
char	16-bit Unicode character
decimal	128-bit precise decimal values with 28-29 significant digits
double	64-bit double-precision floating point
float	32-bit single-precision floating point
int	32-bit signed integer
long	64-bit signed integer
object	Base type for all other types
sbyte	8-bit signed integer
short	16-bit signed integer
string	String value
uint	32-bit unsigned integer
ulong	64-bit unsigned integer
ushort	16-bit unsigned integer

Arrays

```
int[] array = new int[] {1, 2, 3}
int[] array = {1, 2, 3}
var array = new int[] {1, 2, 3}
int[] array = new int[3]
```

Classes

Class	public class Dog {...}	
Inheritance	public class Dog: Pet {...}	
Constructor (no parameters)	public Dog () {...}	Constructors can co-exist
Constructor (one parameter)	public Dog (string nm) {...}	Constructors can co-exist
Field	public string name	
Static Class	public static class Dog {...}	Must only have static members
Static Member	public static int = 1	
Finalizer (destructor)	~Dog () {...}	Cannot have modifiers or parameters

Naming Conventions

Class	MyClass
Method	MyMethod
Local variable	myLocal/Variable
Private variable	_myPrivateVariable
Constant	MyConstant

Logical Operators

and	&&
or	
not	!
xor	^

Arithmetic Operators

+	Add numbers
-	Subtract numbers
*	Multiply numbers
/	Divide numbers
%	Compute remainder of division of numbers

Statements

if-else	if (t) {...} else if (t) {...} else {...}
switch	switch(var) { case 1: break; default: break; }
for	for (int i=10; i<10; i++) {...}
foreach-in	foreach (int i in array) {...}
while	while (t) {...}
do... while	do {...} while (t);
try-catch-finally	try {...} catch (MyException e) {...} catch {...} finally {...}

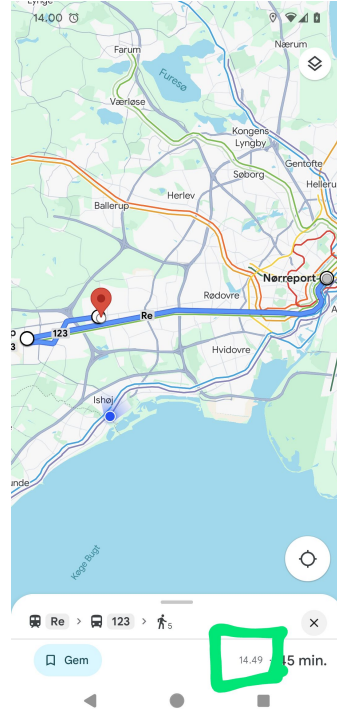
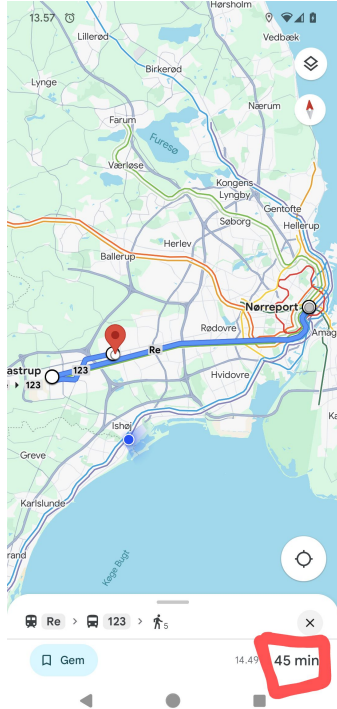
Access Modifiers

public	Accessible by any other code in the same assembly or another assembly that references it
private	Only accessible by code in the same class or struct



Vi skal vise

Google maps



Vi skal udregne:

- 1) Rejsetid**
- 2) Ankomst tid**

Feature nr. 1

1)

Rejsetid

Kode uden metoder/funktioner

```
Int minutes_in_train = 20;
```

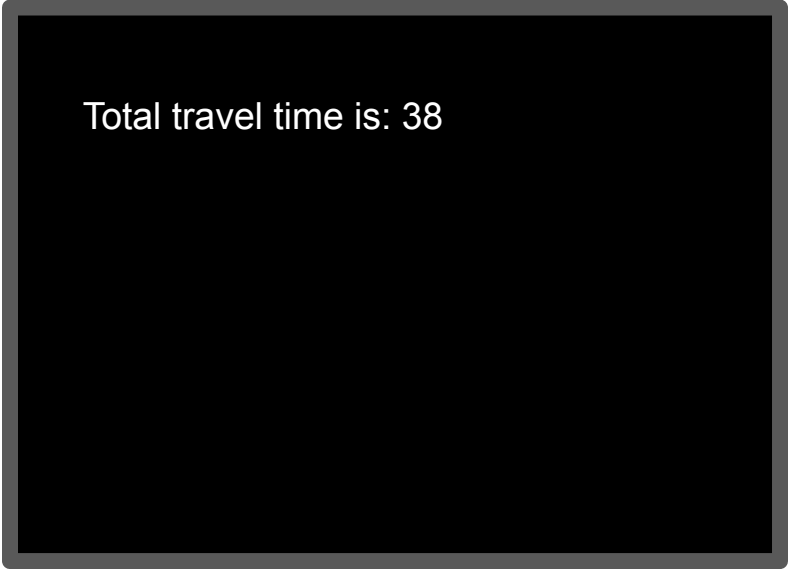
```
Int minutes_in_bus = 10;
```

```
minutes_in_train + 5;
```

```
minutes_in_bus + 3;
```

```
System.Out.Print("Total travel time is: "  
minutes_in_train + minutes_in_bus);
```

Consol:



Total travel time is: 38

God kode er
let at forstå for andre

- A) “No magic numbers”
- B) God navngivning
- C) Minimum gentagelser

“No magic numbers”

```
Int minutes_in_train = 20;
```

```
Int minutes_in_bus = 10;
```

```
minutes_in_train = minutes_in_train + 5;
```

```
minutes_in_bus = minutes_in_bus + 3;
```

```
System.Out.Print("Total travel time is: "  
minutes_in_train + minutes_in_bus);
```

```
Int minutes_in_train = 20;
```

```
Int minutes_in_bus = 10;
```

```
Int minutes_delay = 5;
```

```
minutes_in_train = minutes_in_train +  
minutes_delay;
```

```
Int minutes_delay = 3;
```

```
minutes_in_bus = minutes_in_bus + minutes_delay;
```

```
System.Out.Print("Total travel time is: "  
minutes_in_train + minutes_in_bus);
```




Find 5 fejl

Stavefejl løses med variabel

Minimum gentagelser

```
Int minutes_in_train = 20;
```

```
Int minutes_in_bus = 10;
```

```
static void TrainIsDelayed()
```

```
{
```

```
    minutes_in_train + 5;
```

```
}
```

```
static void BusIsDelayed()
```

```
{
```

```
    minutes_in_bus + 3;
```

```
}
```

```
System.Out.Print(Total travel time is: minutes_in_train +  
minutes_in_bus);
```

Opgaver med variabler og metoder/funktioner

I)

II)

Du mangler lagerplads på din computer og vil ryde op i dine filer.

Lav et program der fjerner alle dupliserede filer med samme navn og filformat.

Næste gang . . .

- Clas
- Objekt orienteret programmering

Feature 1: Lav UI

Fremgangsmåde:

Trin 1) Download Visual studio 2022 Community

Trin 2) Åben programmet “Visual studio installer”

Trin 3) Vælg “.NET desktop development” og tryk install / installer

Trin 4) Åben programmet “Visual studio 2022 community”

Trin 2) Lav nyt projekt

Trin 3) Vælg C# .NET Winforms

... Visual studio genererer nu en masse kode til dig ...

Trin 4) Tryk på play knap for at tjekke at alt virker.

